

TECHNICAL TRAINING PROGRAM

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Network
Infrastructure

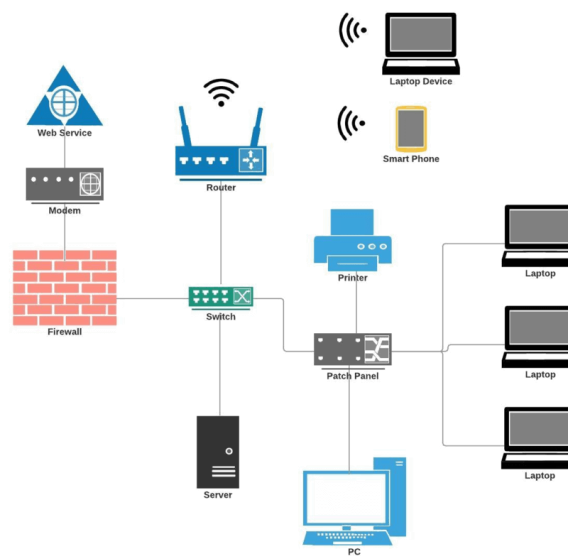
Chapter 5 – Networking Fundamentals

What is a Network

It is useful to distinguish between a **physical device** and a **network node**.

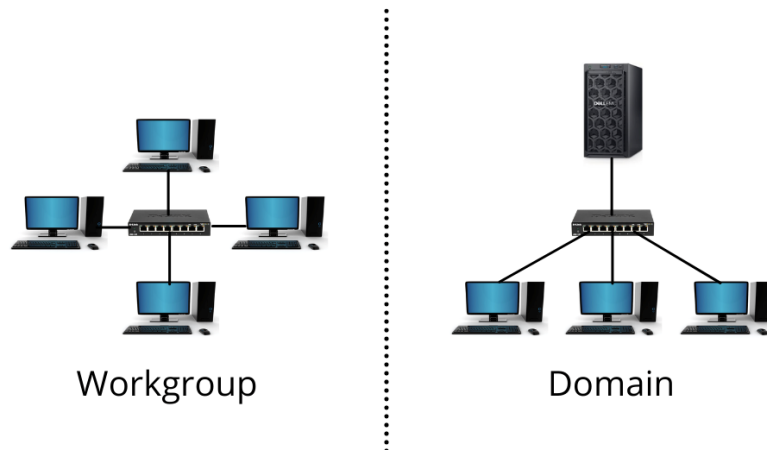
A network node is any physical or virtual endpoint or intermediary that participates in network communication. Examples include computers, servers, printers, routers, switches, Virtual Machines, and virtual network appliances.

In this material, the term physical device refers specifically to hardware, while a network node may be physical or virtual.



A Network Diagram

Workgroup vs Domain



[Workgroup vs Domain](#)

A **Windows Workgroup** is a *peer-to-peer* arrangement in which each computer manages its own local user accounts, security settings, and shared resources.

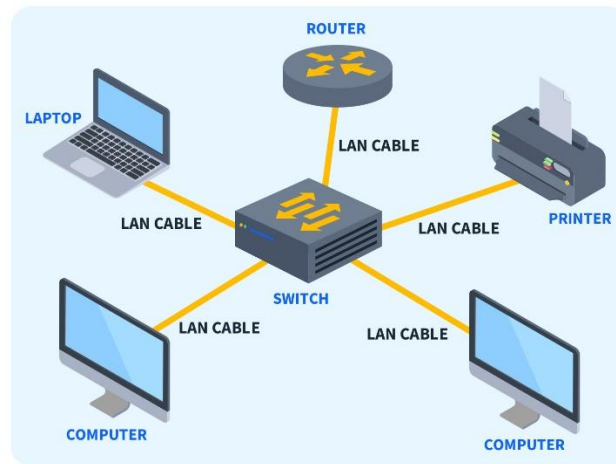
No computer centrally controls the other computers in the Workgroup. Access to a shared folder, printer, or other resource depends on the accounts and permissions configured on the computer that provides the resource.

A **Windows Domain** is a *centrally managed environment* in which user accounts, computers, security policies, and access to network resources can be administered through services such as Active Directory Domain Services.

Domains are commonly used in organizations because they provide centralized authentication, authorization, policy management, and administration.

Network Types

Local Area Network (LAN)



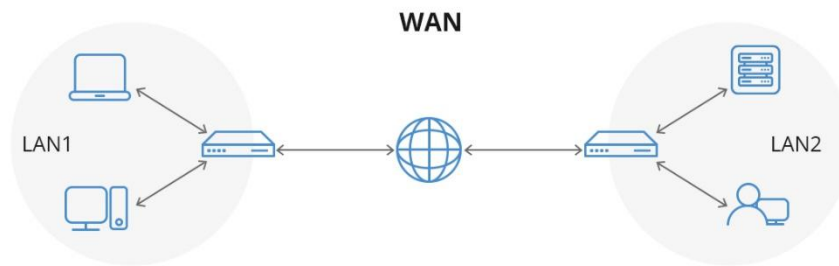
A LAN Network

A *Local Area Network*, or **LAN**, connects devices within a limited geographical area, such as a home, office, school, building, or campus.

A computer used by an employee or another end user may be called a workstation. However, a LAN may also contain servers, printers, switches, routers, Access Points, phones, cameras, and other network devices.

A LAN can operate with or without a dedicated server. Devices may connect through wired Ethernet, wireless technology, or a combination of both.

Wide Area Network (WAN)

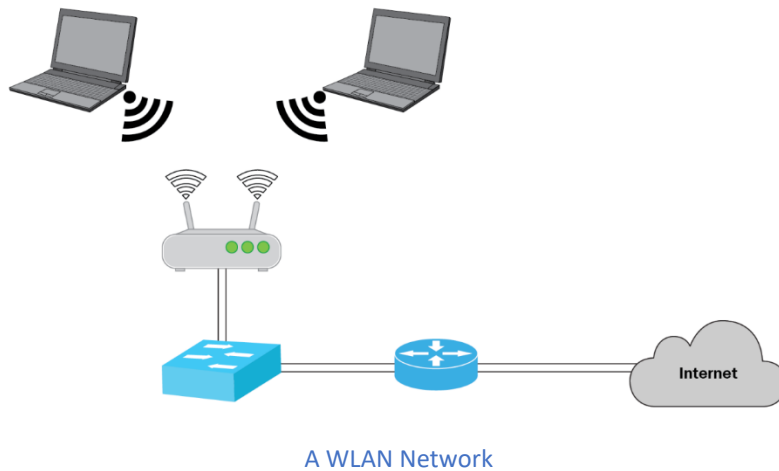


A WAN Network

WAN links often have *higher latency* than LAN connections because traffic may travel over longer distances and pass through service-provider infrastructure.

However, the available bandwidth, reliability, and overall performance depend on the type of WAN service, the provider's infrastructure, the geographical distance, and the network design. Modern WAN services can provide very high transmission speeds.

Wireless Local Area Network (WLAN)

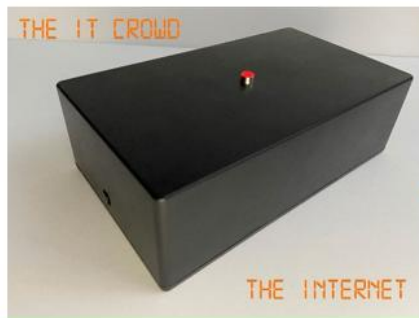


A Wireless Local Area Network, or WLAN, is a LAN that uses wireless technology, normally Wi-Fi, to connect compatible devices. Wireless clients commonly connect through an *Access Point*. The Access Point normally bridges wireless devices to the wired LAN.

Internet access, when available, is normally provided through a router or another Default Gateway. A WLAN can exist and support local communication even when no Internet connection is available.

One major advantage of wireless networking is mobility. Users can move within the coverage area while maintaining their network connection.

Internet



This is **NOT** the Internet



This is the Internet

The Internet is a global system of interconnected networks that communicate by using the TCP/IP protocol suite.

It depends on physical infrastructure such as routers, switches, servers, data centers, copper and fiber-optic cables, wireless links, and submarine cables. However, it also depends on logical addressing, routing, protocols, and services that allow independent networks to operate together.

For this reason, the Internet is commonly described as a network of networks.

The Internet supports many services such as:

- **World Wide Web**: it's a library consisting of files, documents, videos, pictures etc. that can be accessed over the internet.
- **eMail**: It's a service that let us send and receive messages over the Internet.
- **Social Media**: They are online platforms/applications that allow users to interact with each other.
- **Search Engines**: It's the tool that helps us to find information on the Internet.



Intranet – Extranet

Intranet

An **Intranet** is a private network or collection of internal services used by an organization.

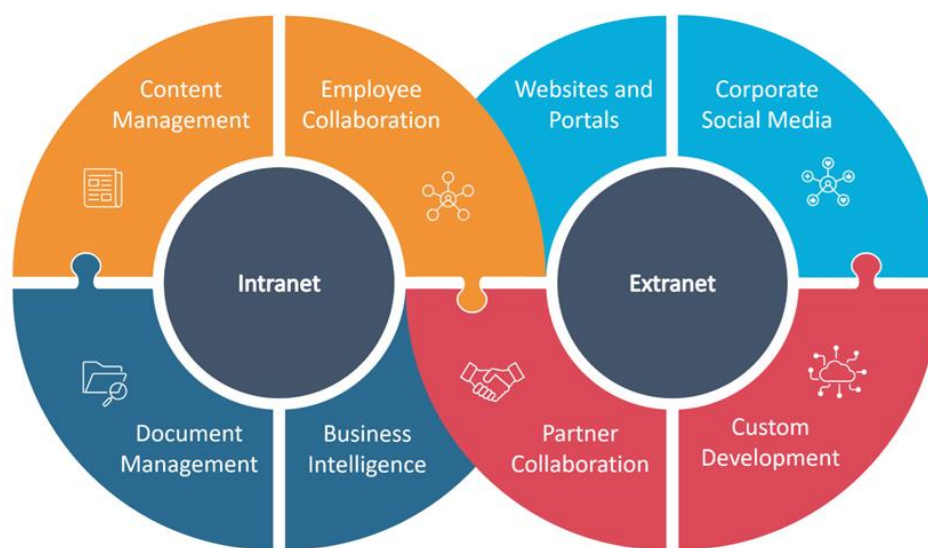
It allows authorized users to access internal information, applications, documents, and other organizational resources. Access may be controlled through authentication, permissions, network-security controls, and organizational policies.

An Intranet may or may not provide access to the public Internet.

Extranet

An Extranet is a controlled extension of an organization's internal environment that provides selected external users, such as customers, suppliers, or business partners, with access to specific resources. Access may be provided through a **VPN**, a secure web portal, an authenticated application, or another protected access method.

External users should receive access only to the resources required for their approved purpose.



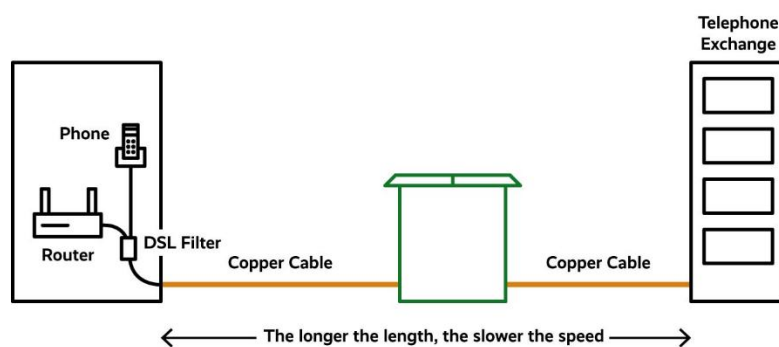
Intranet + Extranet

Data Transmission Technologies

Asymmetric Digital Subscriber Line (ADSL)

ADSL and ADSL2+ transmit data through existing copper telephone lines. They are asymmetric technologies, meaning that the downstream and upstream speeds are different.

ADSL2+ can provide better performance than earlier ADSL implementations. However, DSL technologies are increasingly considered legacy access solutions, although they may still be used in locations where newer infrastructure is unavailable.

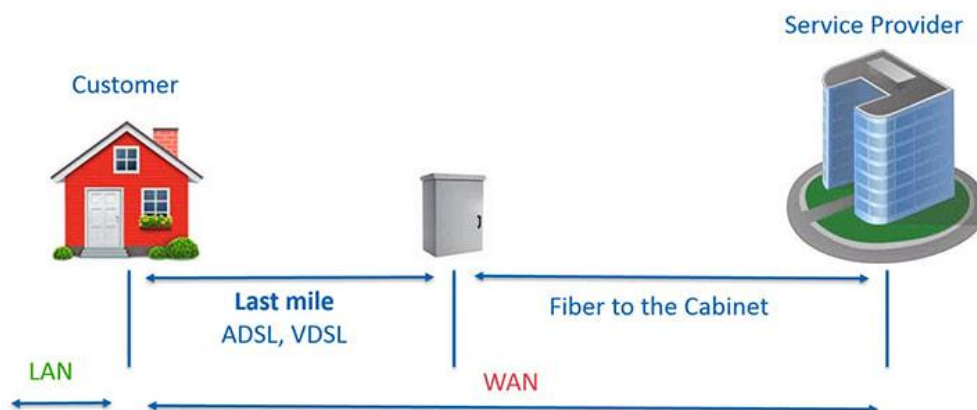


ADSL Diagram

Very high-speed Digital Subscriber Line (VDSL)

VDSL and VDSL2 also use copper telephone infrastructure but can provide higher data rates than ADSL over suitable line conditions.

Performance is strongly affected by the distance between the customer and the provider's equipment, the quality of the copper line, electromagnetic interference, crosstalk, and the technology profile being used. Performance normally decreases as the copper-line distance increases.

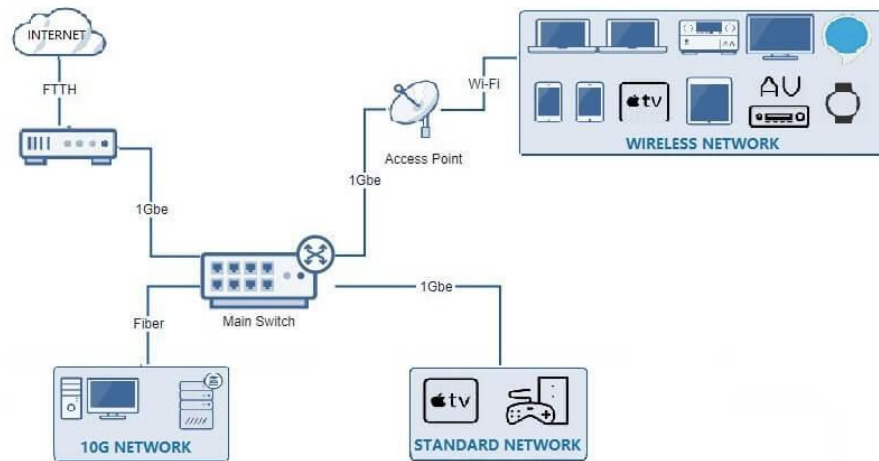


VDSL Diagram

Fiber Optics

Fiber-optic technology transmits data by using modulated light signals through glass or plastic optical fibers.

It can provide high bandwidth, low signal attenuation over long distances, and strong resistance to electromagnetic interference. The achievable distance and speed depend on the fiber type, optical transceivers, wavelength, and network standard.



A network consisting of copper cables, Wi-Fi, and Fiber Optics

Network Topologies

We use the term Network Topology to describe the way that the various network devices are connected and arranged within a communication network. This includes both the physical layout, which refers to the actual placement and connections of hardware devices (such as computers, switches, and routers), and the logical layout, which relates to the path that data must follow as it crosses the network.

The most common types of Network Topology are:

Point – to – Point Topology

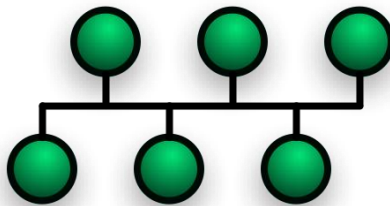
It is the direct connection between two nodes or devices. It is the simplest type of network topology as it has only two endpoints?



Point – to - Point Topology

Bus Topology

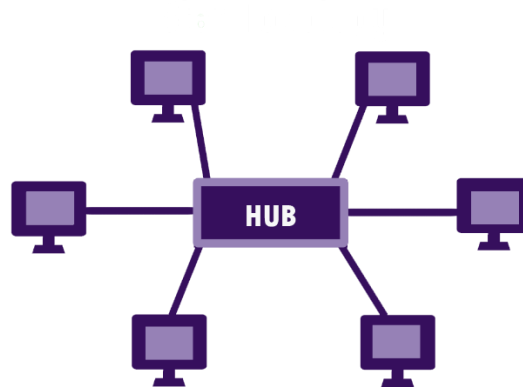
In this topology all devices are connected to a single central cable which is called backbone.



Bus Topology

Star Topology

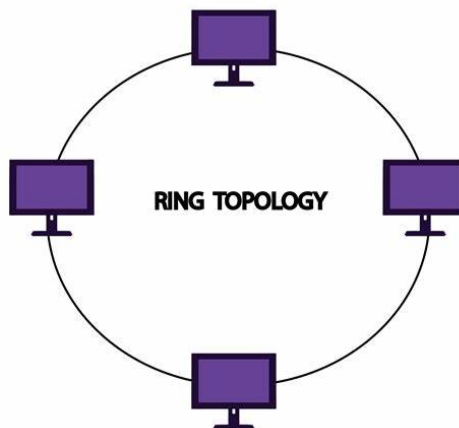
It's the most common topology of all. Here all devices are connected to a central hub or switch, which acts as the central communication channel for data transmission.



Star Topology

Ring Topology

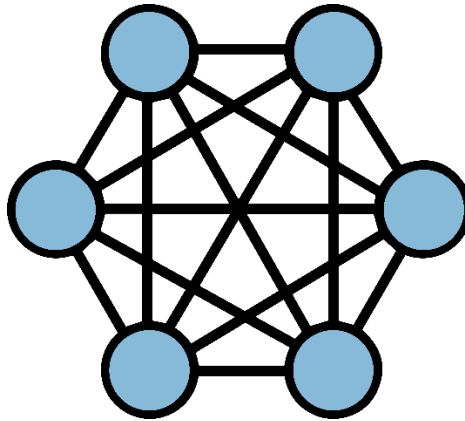
In this topology, every node is connected to two other nodes forming a circular path for the data, similar to a ring.



Ring Topology

(Full) Mesh Topology

In this type of topology, each node is connected to every other node. This creates multiple paths for data to travel in the network, on the one hand enhancing redundancy but on the other hand raises the cost.

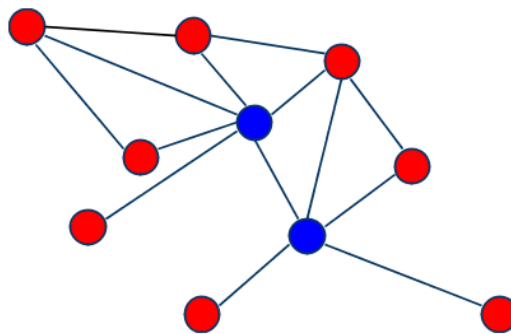


Full Mesh Topology

Partial Mesh Topology

In a *Partial Mesh Topology*, some nodes have connections to multiple other nodes, but not every node is directly connected to every other node.

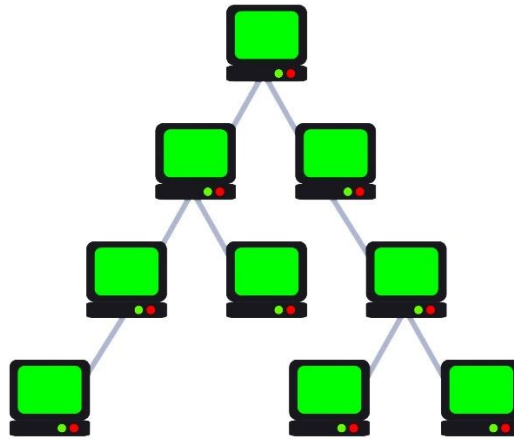
This provides some alternative communication paths while requiring fewer links than a Full Mesh Topology.



Half Mesh Topology

Tree Topology

This specific topology is a combination of Bus and Star Topologies. The whole structure is organized in hierarchical manner, like a tree.



Tree Topology

Protocols and Protocol Suite

Network communication is governed by agreed rules called protocols. A protocol defines how information is *formatted, addressed, transmitted, received, and interpreted*. It may also define timing, sequencing, error handling, and the actions that communicating systems must perform.

Different protocols support different functions. For example, FTP is one protocol that may be used to transfer files between systems, although other file-transfer methods also exist.

A protocol suite is a collection of related protocols that work together to support communication between systems. A protocol suite does not guarantee that every communication will be error-free; reliability depends on the protocols and mechanisms being used.

In the field of IT there are two major implementation models. The TCP/IP (Transmission Control Protocol/Internet Protocol) model and the OSI (Open Systems Interconnection) model. Very important to know is that we implement TCP/IP rules, but when we have to troubleshoot or refer to a specific portion of the IT infrastructure, we follow the OSI terms.

The network IT infrastructure is divided in Layers. The OSI model, which is also called as ***OSI Reference model***, it consists of seven layers and is far more analytic than TCP/IP which has only four Layers.

OSI Reference Model	TCP/IP Model	TCP/IP Model <i>new</i>
Application Layer	Application Layer	Application Layer
Presentation Layer		
Session Layer		
Transport Layer	Transport Layer	Transport Layer
Network Layer	Internet Layer	Network Layer
Data Link Layer	Link Layer	Data Link Layer
Physical Layer		Physical Layer

OSI vs TCP/IP vs TCP/IP *new*

The Functions of the Seven Levels

Layer Seven (Application Layer)	Enables Users And Applications To Access Network Services
Layer Six (Presentation Layer)	Translates Data Into A Common Format
Layer Five (Session Layer)	Establishes A Communication Session Between Applications
Layer Four (Transport Layer)	Manages Message Fragmentation And Reassembly
Layer Three (Network Layer)	Manages Logical Addressing, Data Routing Creating Subnetworks
Layer Two (Data Link Layer)	Provides local-link frame delivery, MAC addressing, media access, and frame-error detection
Layer One (Physical Layer)	Physical Network Media And Signal Methods

Note: Network functions are associated with different OSI layers, but the distinction between hardware and software is not absolute.

Modern network devices may use hardware, firmware, and software to process information from multiple layers, including Layers 4 through 7.

Protocol Data Units PDUs

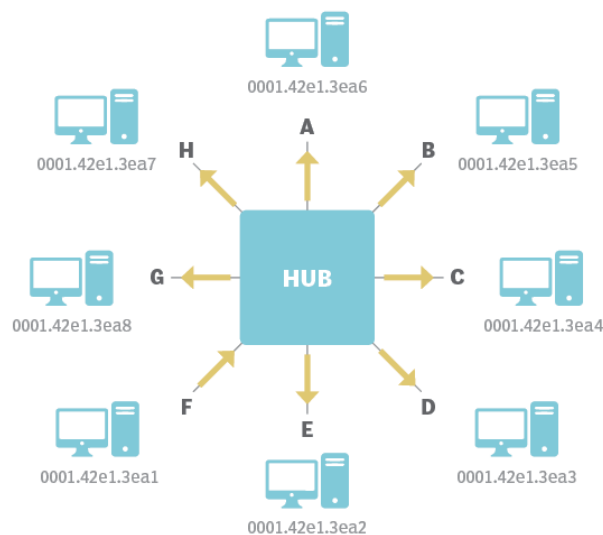
OSI Layer Protocol Data Units	
Transport Layer	Segment
Network Layer	Packet
Data Link Layer	Frame
Physical Layer	Bits

Network Devices

Hub (Layer 1 device)

The **HUB** is an older network device that is rarely used in modern networks, it is used to connect many network devices such as computers printers within a local LAN network. Its mode of operation was quite simple, when a data packet was arriving at one of its ports, the HUB forwarded (Broadcasted) the traffic out of all its remaining **active** ports, without examining the packet for errors, checking the sender or the receiver. This means that all devices connected to the hub would receive and process, up to a specific level, the packet regardless of who the recipient was.

This caused major problems and delays in communication because all devices connected to the Hub were part of a Collision Domain. That is, if two devices were sending data at the same time, there was a high probability that conflicts would occur during transmission.



How the Hub Handles Traffic

To manage to mitigate the problem they introduced a mechanism new for the time, **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection).

But how did this mechanism work?

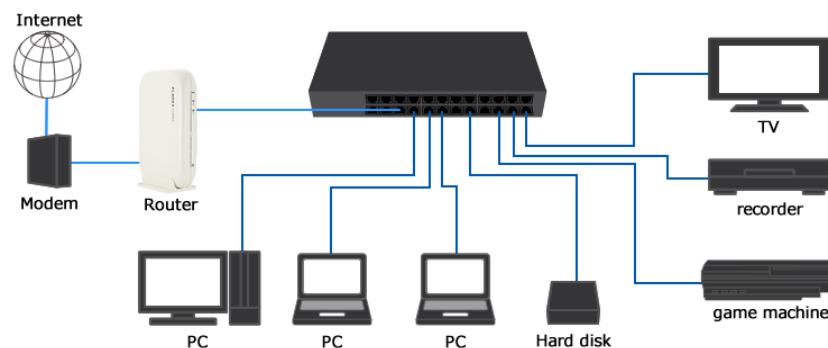
- The sending device was monitoring the transmission medium to see if it was idle and so to begin to send data.
- Multiple devices can access the medium but only one can send data at a time.
- When two devices were sending data in parallel then a collision would occur. The devices would then detect the collision and stop the transmission.
- Finally, after the collision, a random clock was run for each device before trying to send data again.

The collision problem has been minimized by the extensive use of switches in modern networks. A Switch creates a separate collision domain for each connected device.

Switch (Layer 2 device & Layer 3 device)

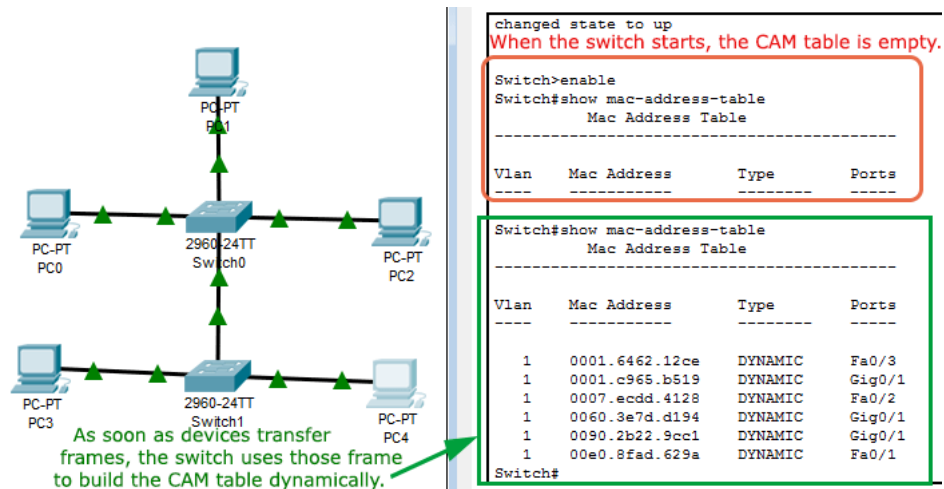
A switch is a very basic network device. It is used like the hub device to connect several computers or other network devices to a local network. Its main job is to receive a data frame from the sender and forward it to the appropriate final recipient. However, the operation mode of the switch is completely different from that of the hub. When a data packet arrives at the switch port, after checking some specific characteristics of the packet, it forwards it to the specific port to which the recipient is connected.

The switch checks the MAC Address of the frame and compares it with the MAC Address Table stored internally. So, it can check if there is a record with the recipient's details and forward the packet out the appropriate port.



Network Devices Connected on a Switch

In order for a switch to build its internal MAC Address Table, it must follow a process called MAC Learning. So, when a Switch receives an Ethernet frame on one of its ports, it examines the MAC Address of the frame and if it finds that there is no corresponding entry, it registers it together with the number of the port that received the frame. Each of these records has a particular lifetime (*Aging Time*). If the switch does not receive any traffic from a specific MAC within a certain time, then this record is deleted. It is a basic process of the Switch, which allows it to forward packets only to the appropriate ports and thus manages to reduce unnecessary traffic within the network.



CAM (Content Addressable Memory) Table – MAC Table

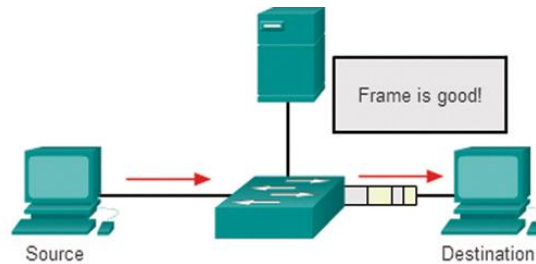
The Flooding can occur in some cases such as:

- ◆ In case that a switch receives a frame with an unknown destination MAC Address, it Broadcasts the specific frame to all its active ports and waits to receive a response from the appropriate recipient in order to complete the MAC Address Table.
- ◆ When it is fresh installed and has to learn the devices, it has connected to it.

Switches can operate in three different modes to forward traffic

i. Store and Forward

In this mode the switch stores temporarily the whole frame and investigates the Frame Check Sequence, and if it doesn't find any problem it forwards it the receiver. In case there is an error, the switch drops the frame.

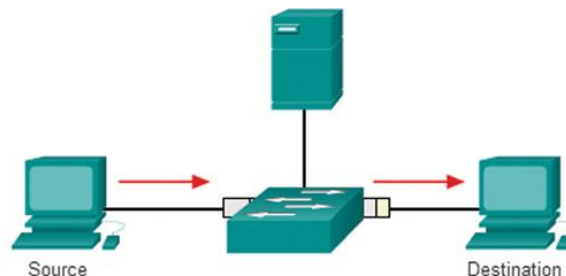


A store-and-forward switch receives the entire frame, and computes the CRC. If the CRC is valid, the switch looks up the destination address, which determines the outgoing interface. The frame is then forwarded out the correct port.

Store n Forward Method

ii. Cut Through

Different mode of operation from Store and Forward, and considerably faster. In this mode the switch only examines the recipient's MAC Address and starts forwarding the frame to it without waiting to receive the entire message.



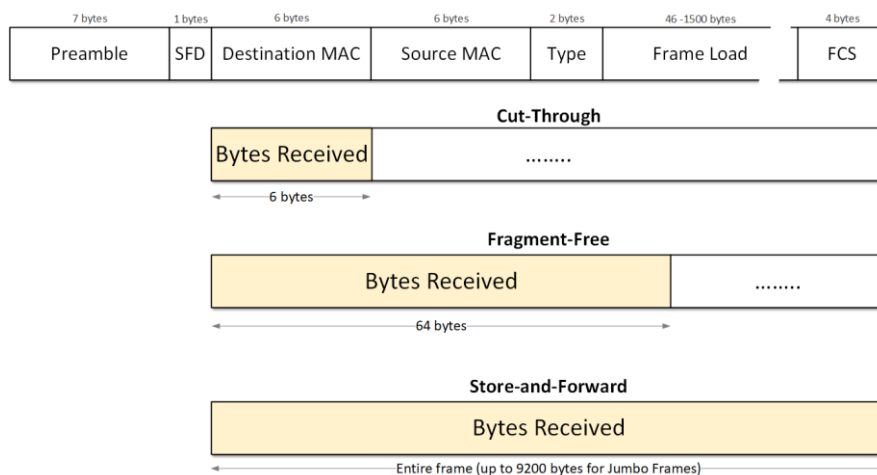
A cut-through switch forwards the frame before it is entirely received. At a minimum, the destination address of the frame must be read before the frame can be forwarded.

Cut-Through Method

iii. Fragment-Free

This case is a combination of the previous two functions. The switch waits to receive the first 64 bytes of the frame before it starts forwarding traffic. This serves to avoid *Collision Fragments* which usually occur in the first 64 bytes of a message.

Each of these three modes have advantages and disadvantages, the choice really depends on the specific network requirements such as the need for high data integrity or it is important for the network to have low latency.



Switching Methods

Switching Methods Pros & Cons

Store and Forward	
Pros	Cons
Error Checking	Higher Latency
Security	Resource Intensive
QoS	

Cut Through	
Pros	Cons
Low Latency	Error Spread
Speed	Limited Error Handling
Efficiency	Specific Use Cases

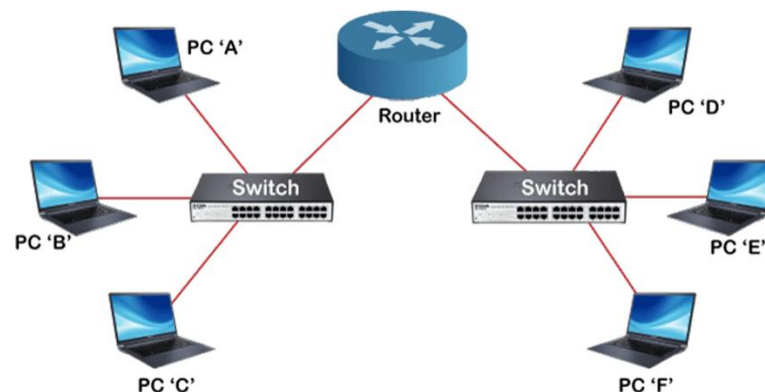
Fragment Free	
Pros	Cons
Reduced Error Propagation	Partial Error Checking
Balanced Latency	Moderate Latency
Improved Network Performance	Specific Use Cases



Router (Layer 3 Device)

A router is a network device that is responsible for sending and receiving data between different networks. It chooses to which network and through which path to send the data and joins different networks and subnets. Within a LAN, if a device wants to communicate with a different network than its own, such as the Internet, then all it needs to know is the “*Default Gateway*” of the network it belongs to.

Another basic function of the Router is to divide the network into “*Broadcast Domains*”. Each router port is also a separate Broadcast Domain. A Broadcast Domain includes all network devices that can communicate with each other at OSI Layer 2 via broadcasting.



One Router - Two Networks

The router, like the switches, creates an address table called “*Routing Table*”. This table contains all the IP addresses of the networks that the router knows, either statically (Manually) or dynamically (Routing Protocol), so that it can route the traffic.

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

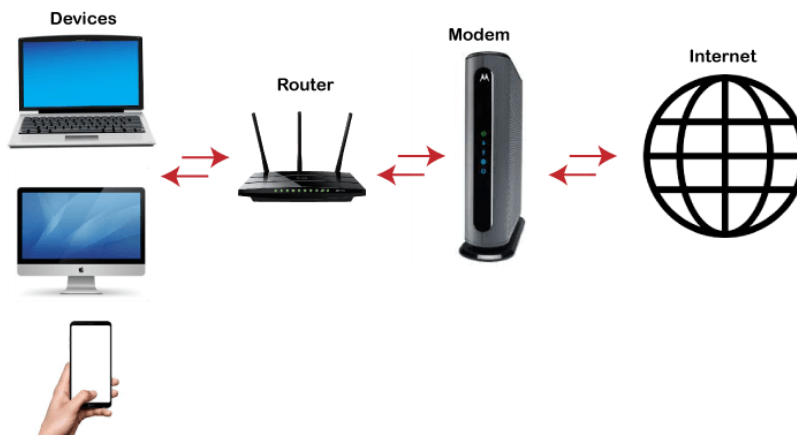
Gateway of last resort is 192.168.0.5 to network 0.0.0.0

R    10.0.0.0/8 [120/1] via 192.168.0.10, 00:00:05, GigabitEthernet0/2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, GigabitEthernet0/0
L    20.0.0.1/32 is directly connected, GigabitEthernet0/0
R    30.0.0.0/8 [120/1] via 192.168.0.5, 00:00:05, GigabitEthernet0/1
    192.168.0.0/24 is variably subnetted, 5 subnets, 2 masks
R    192.168.0.0/30 [120/1] via 192.168.0.5, 00:00:05, GigabitEthernet0/1
    [120/1] via 192.168.0.10, 00:00:05, GigabitEthernet0/2
C    192.168.0.4/30 is directly connected, GigabitEthernet0/1
L    192.168.0.6/32 is directly connected, GigabitEthernet0/1
C    192.168.0.8/30 is directly connected, GigabitEthernet0/2
L    192.168.0.9/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 192.168.0.5
```

Routing Table

Modem

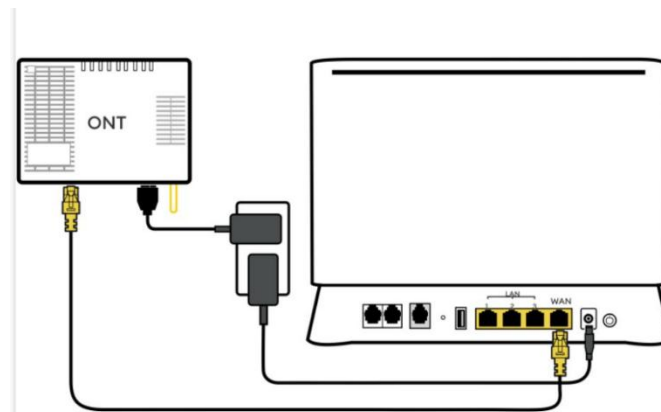
The Modem (**M**odulator / **D**emodulator) is the device through which we manage to access the Internet. It allows the conversion of Digital Signals from a computer to Analog Signals that way the data can be transmitted over telephone lines or other analog media and vice versa.



Modem

In a Fiber Optic network the modem has been replaced by ONTs (Optical Network Terminal).

An **ONT** is a device used in fiber optic networks to connect a business to the ISP. The ONT converts the optical signals transmitted through the optical fiber into electrical signals that can be used by network devices into our internal network.



ONT Installation

Cabling

In networks, data transfer is either wired or wireless. In this paragraph we will talk about wired networks. The most common and widely used cables are twisted pair copper cables. The cables are twisted to neutralize electromagnetic interference (Crosstalk).

Cable Types

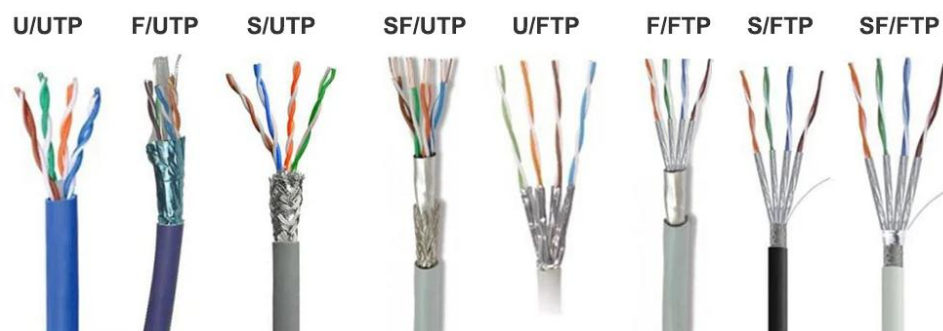
Unshielded Twisted Pair (UTP) cable: It consists of four pairs of cables. It comes in two forms, one is rigid because it has a plastic core inside, and so it goes more easily through channels and walls. There is also the classic cable which is flexible and only has copper wires inside. Each pair has one colored wire and one with the corresponding color but with a white stripe. Depending on the company that produces them, the cables are inside a PVC plastic. Their termination is always on an RJ-45 connector.

Shielded Twisted Pair (STP) cable: Similar to UTP but with additional shielding to further reduce EMI, used in environments with high electromagnetic interference.

Foil Shielded Twisted Pair (FTP) cable: Uses a thin layer of aluminum foil to provide 100% coverage of the conductors it surrounds.

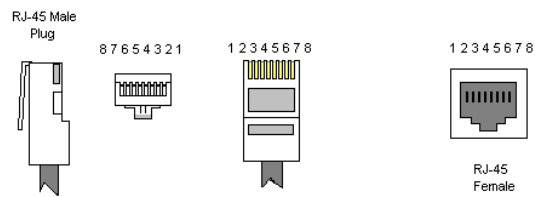
Braided Shielded cable: Uses a mesh of woven copper wires to provide shielding. Offers flexibility and durability.

Of course, there are other types of shielded cables which are a combination of the others such as: S/FTP, F/FTP, SF/FTP cables.



Cable Types

All shielded twisted pair cables they are not using a plastic RJ-45 connector, but a metallic one which provides for better shielding against EMI, are more durable than plastic one and they are more resistant to the heat. Also, although this concerns electricians, in order to get the desired results, there should be proper grounding in all the wiring, even in the sockets.



RJ-45 Connector Pin Numbering



RJ-45 Passthrough



Crimping Tool

Cable Termination

In networking there are two cable termination standards, *T568A* & *T568B*. Both have differences in Pin configuration:

- ❖ **T568A**: White/Green, Solid Green, White/Orange, Solid Blue, White/Blue, Solid Orange, White/Brown, Solid Brown.
- ❖ **T568B**: White/Orange, Solid Orange, White/Green, Solid Blue, White/Blue, Solid Green, White Brown, Solid Brown.

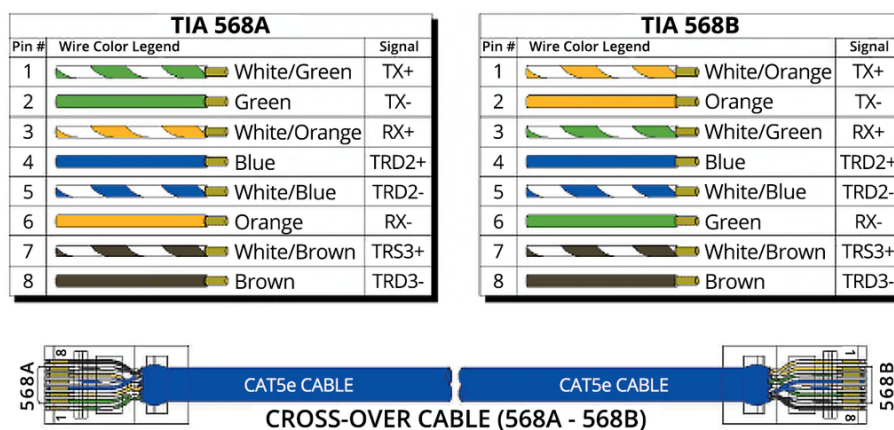
We can use either layout we want but we should follow the same pattern throughout the wiring to avoid mistakes.

There are two Ethernet cable types:

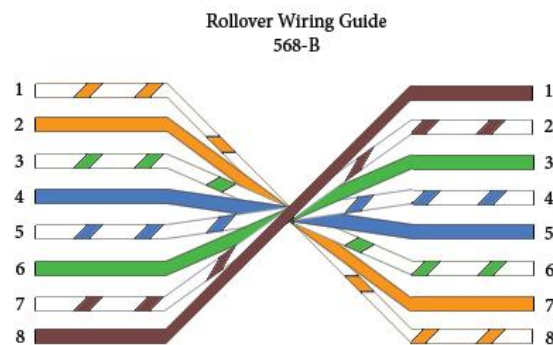
- a. **Straight-Through cable**: When both ethernet cable ends have the same pin configuration then the cable is called Straight-Through and is used to connect different devices such as: Computer to switch, computer to router, router to switch, switch to hub.
- b. **Crossover cable**: In this situation the two cable ends have different pin configuration and is used to connect same devices such as: Computer to computer, switch to switch, router to router, computer to router (consider to be same devices).

In modern implementations, we can use any cable, because in the network cards of the devices we can activate the Auto Sensing function.

The different end configuration is:



There is also another type of cable type which is called **Rollover cable** (*Console cable*) and is used when we need to connect a computer to a router's console port. As we can see below rollover cables have opposite Pin assignments on each end of the cable.



An example of a Rollover cable in 568-B

Fiber Optics Cables

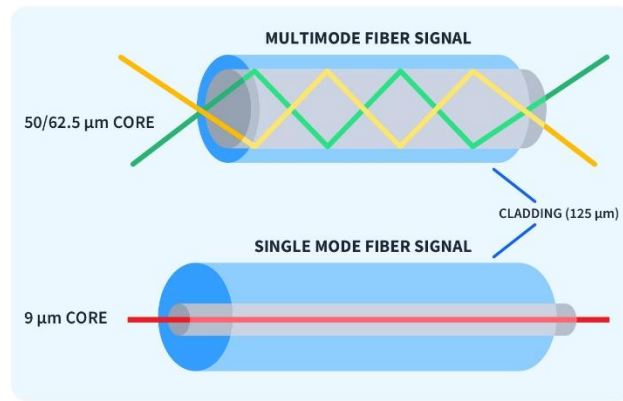
Fiber optics is a cutting-edge technology used for transmitting data in the form of light signals. Using materials such as glass or plastic, FO allow for fast and reliable transmission of large amount of data over long distances.

For this reason we have different types of Fibers

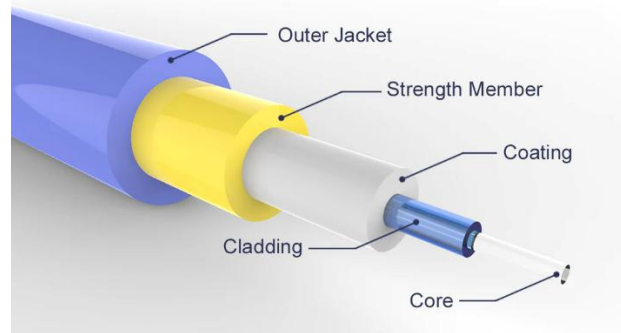
- ☞ **Single-Mode Fiber (SMF)**: We use this type when we need to transmit data over long distances. They have small core and are ideal for telecommunications and WAN implementations.
- ☞ **Multi-Mode Fiber (MMF)**: Different kind of cable with bigger core, and they are able to transmit multiple light paths. They are suitable for short distances and is very common use in LANs, buildings, campuses and connecting equipment like Servers and Switches.



Single-Mode vs Multi-Mode Fiber



Data Transmission in a Fiber Optic Cable



Fiber Optic Cable Layers

OSI Model Layers

Layer 1: Physical Layer

It deals with the physical connection between devices. As we have already mentioned the three first layers are containing hardware components also. So, in L1 we can find cables, hubs, and other devices.

Layer 2: Data Link Layer

Provides node-to-node data transfer and manages error detection and correction from the physical layer. L2 hardware components: Switches, network interface cards (NIC). Protocols: Ethernet, PPP (Point-to-Point Protocol).

In Ethernet, the Data Link Layer uses the Frame Check Sequence to detect frame corruption.

If a frame is detected as corrupted, it is normally discarded. Ethernet does **not** correct the damaged frame or automatically request its retransmission.

Recovery, when required, is provided by another protocol or application mechanism.

The Data Link Layer is divided into two sublayers:

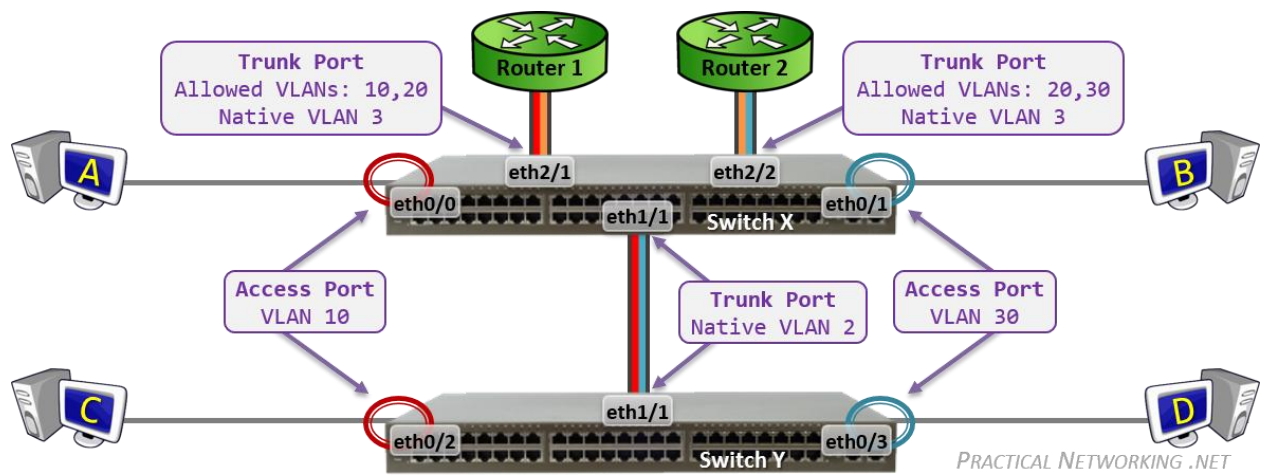
1. **Logical Link Control (LLC)**: It handles error checking and frame synchronization. It provides a way to identify the network layer protocol being used.
2. **Media Access Control (MAC)**: It controls how devices on the network gain access to the medium and permission to transmit data. It deals with *Physical Addressing* and network topology.

Each device within a network has a unique MAC Address which is used for device Identification at L2. MAC Addresses are 48-bit long and are assigned on NICs. The Ethernet protocol is the most important L2 Protocol. It is also known as IEEE 802.3, is a set of rules that enable devices to communicate in a LAN environment.

A very important feature of L2 is **VLAN (Virtual Local Area Network)**, which allows for network segmentation improving network security and performance. In Data Link Layer we can apply the Link Aggregation feature, which combines multiple network connection to increase bandwidth and provide redundancy. Moreover, in L2 we can configure QoS in order to prioritize specific traffic to increase performance.

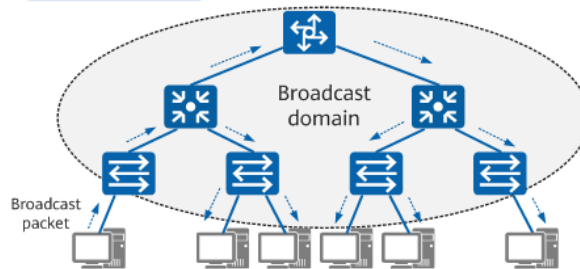
VLAN most important characteristics:

- **Segmentation**: A VLAN can divide a network into smaller ones and thus create separate Broadcast Domains. This means that devices from different VLANs cannot communicate directly to each other without a layer 3 device, like a router or a layer 3 switch.
- **Flexibility**: By using VLANs an administrator is able to group network devices based on function, or by department, regardless of their physical location.
- **Traffic Management**: With VLANs an administrator can isolate the traffic and improve performance by reducing congestion.
- **Security**: VLANs provide logical segmentation and create separate broadcast domains. However, a VLAN alone does not enforce complete access control. Communication between VLANs should be controlled through routing policies, Access Control Lists, firewalls, authentication, and other security controls.
- **VLAN Tagging**: VLANs for the purpose of network traffic identification and management use tagging protocols such as IEEE 802.1Q. This way multiple VLANs can share the same physical medium while remaining logically separate.

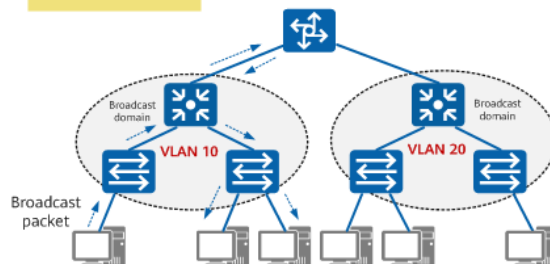


VLAN

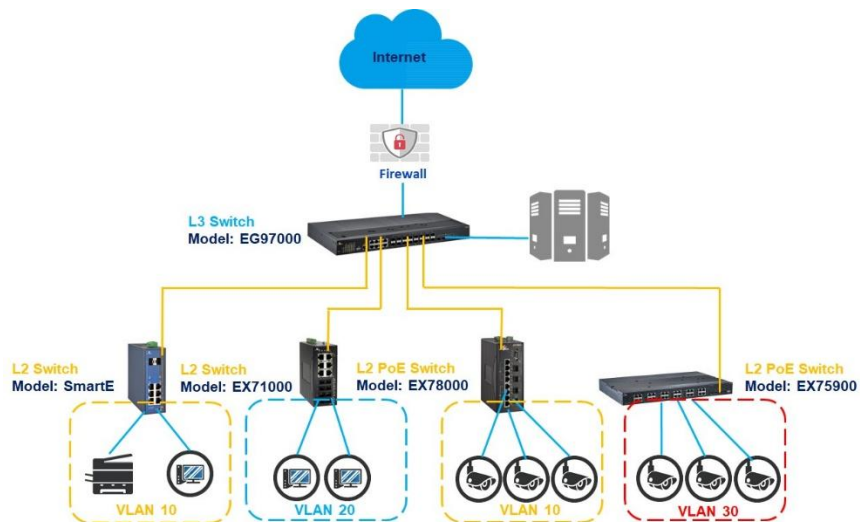
Without VLANs



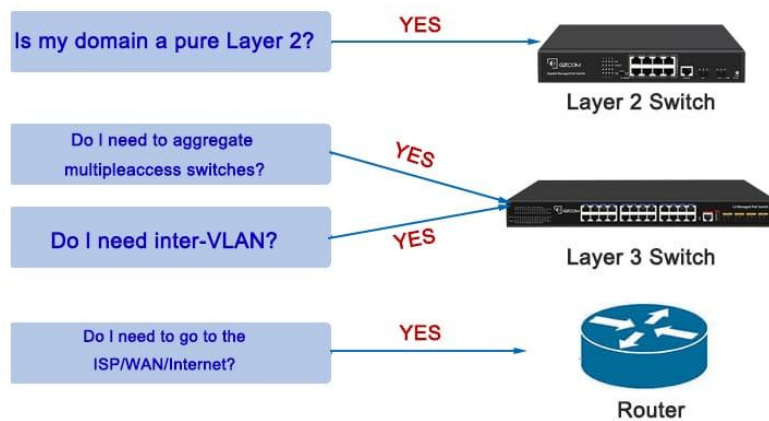
With VLANs



Broadcast traffic With/Without VLANs



A Network Topology



Switch L2 vs Switch L3 vs Router

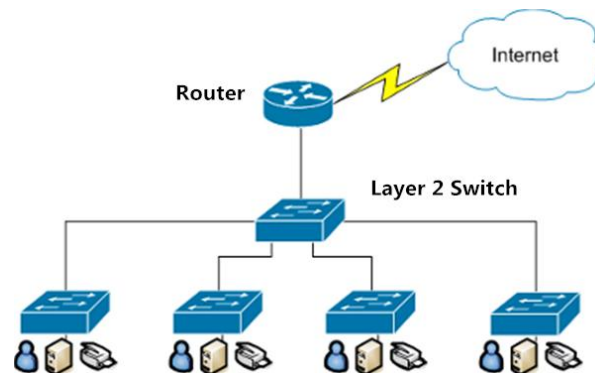
Layer 3: Network Layer

Manages the routing of data packets between devices across different networks. It is a connectionless protocol, which means that the recipient does not know that someone is sending those data and also there is no predetermined path that the data will follow from the sender to the recipient. Finally, the IP protocol is not involved with the proper transport of the packet, which is the responsibility of a higher protocol.

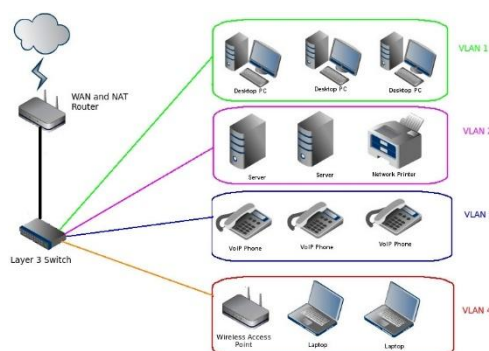
The **Network Layer** deals with the Logical Addressing of the devices The Network Layer uses logical IP addressing to identify interfaces and route packets between networks.

IP addresses may be configured manually or assigned through mechanisms such as DHCP, DHCPv6, or SLAAC (Stateless Address Autoconfiguration). This layer determines the most efficient path for data traffic from source to destination device. Since the path is determined the next thing is to forward traffic from one network to another until the data reach the receiver. Many times, a packet is too large to be transmitted over a network, so it must be break down into smaller fragments to be easier to send it from one destination to another. As soon as the data packet reaches its final destination the Network Layer undertakes to reassemble it into its original form.

Each Network Interface has a unique IP address. There are two versions of IP protocol, the IPv4 and IPv6. Since 2012 IPv4 has been exhausted and we have moved to IPv6, but it is not that often used in internal networks at least.



Router with Layer 2 Switch

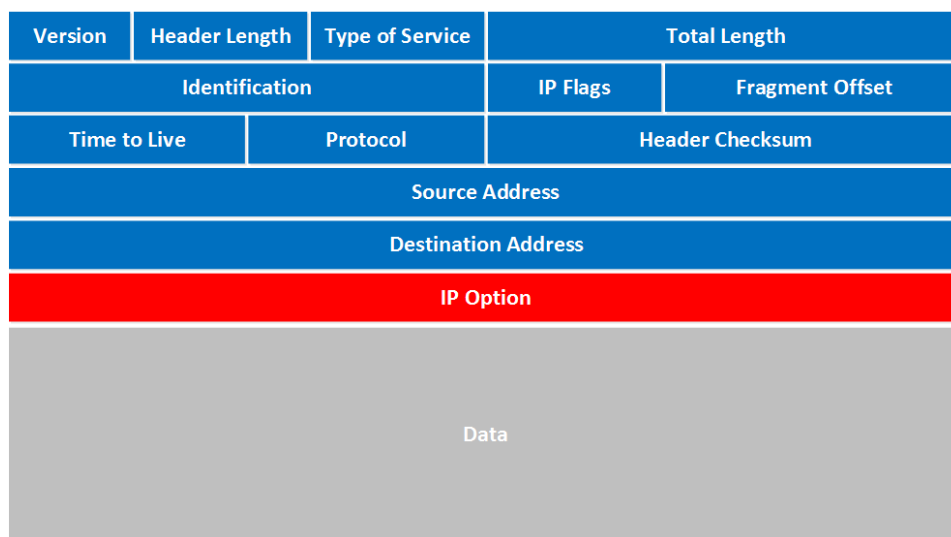


Router with Layer 3 Switch with VLANs configured



IPv4 Key Features

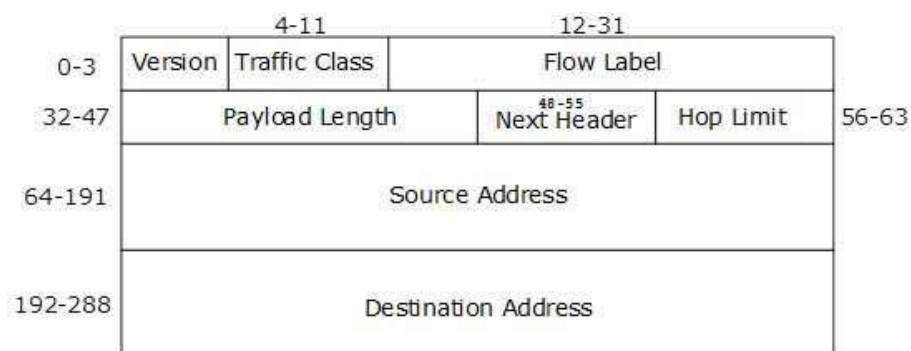
- ◆ Is a 32-bit address displayed as four decimal numbers separated by dots (.) (192.168.1.1) it allows for approximately 4.3 billion unique addresses.
- ◆ Can be configured manually or dynamically through DHCP (Dynamic Host Configuration Protocol).
- ◆ Security is an optional feature and if needed is implemented by IPsec (Internet Protocol Security).
- ◆ Uses ARP (Address Resolution Protocol) to map IPv4 addresses to MAC addresses.
- ◆ Supports QoS features through ToS (Type of Service) field in IP header.



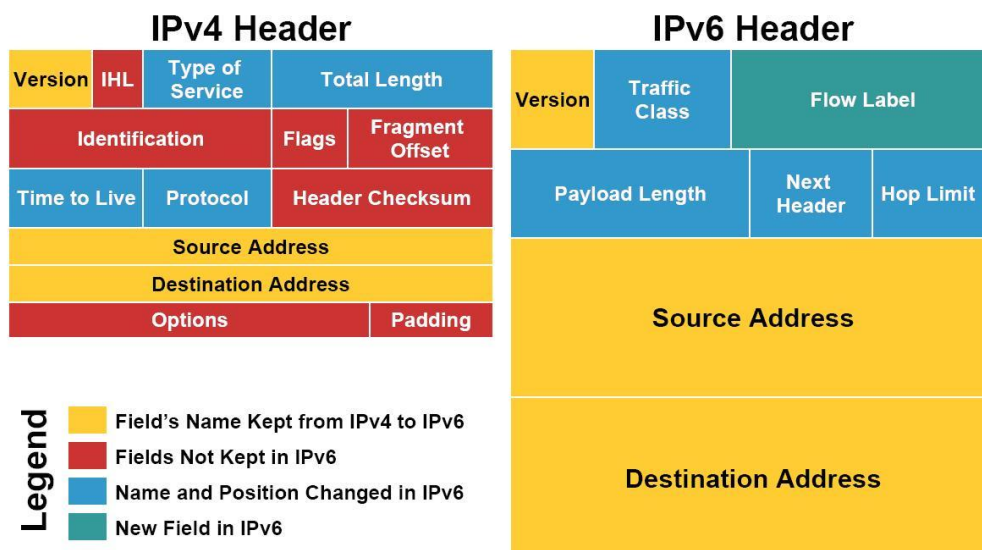
IPv4 Header

IPv6 Key Features

- ◆ Is a 128-bit address displayed as eight groups of four hexadecimal digits separated by colons (:) (2001:0db8:85a3:0000:0000:8a2e:0370:7334), this allows for a huge amount of IPv6 addresses, approximately 340 undecillions.
- ◆ Supports automatic address configuration through SLAAC (StateLess Address AutoConfiguration) and DHCPv6. Of course, we can configure a network interface manually, but it is something that can easily lead to an error.
- ◆ IPv6 was designed with support for IPsec, but IPv6 traffic is not automatically encrypted. Security depends on whether IPsec or another security technology has been configured and used.
- ◆ Uses NDP (Neighbor Discovery Protocol), to map IP addresses to MAC Addresses.
- ◆ It also supports QoS through Flow Label Field.



IPv6 Header



IPv4 vs IPv6

IPv4 Private Addresses

Private IPv4 address, is a range of IP reserved for use within private networks. These addresses are not *routable* on the public internet, meaning they cannot be accessed directly from outside the local network. Instead, they are used for internal communication within a network, such as a home or office LAN.

Private IPv4 Classes

Class A	
Range	10.0.0.0 – 10.255.255.255
Subnet Mask	255.0.0.0 or /8
Usage	Very Large Networks

Class B	
Range	172.16.0.0 – 172.31.255.255
Subnet Mask	255.240.0.0 or /12
Usage	Medium to Large Networks

Class C	
Range	192.168.0.0 – 192.168.255.255
Subnet Mask	255.255.0.0 or /16
Usage	Small to Medium Networks

Public IPv4 Addresses

A public IPv4 address is a unique IP address and is assigned to a device that is connected to the internet. These addresses are *routable* on the Internet, meaning they can be accessed from anywhere in the world. Public IP addresses are crucial for devices that need to communicate over the internet, such as web servers, email servers, and any device that needs to be accessible from outside a local network.

A public IPv4 address is an address that may be globally routed through the Internet.

However, not every IPv4 address outside the RFC 1918 private ranges is a normal public address. IPv4 also contains several special-purpose and reserved address ranges.

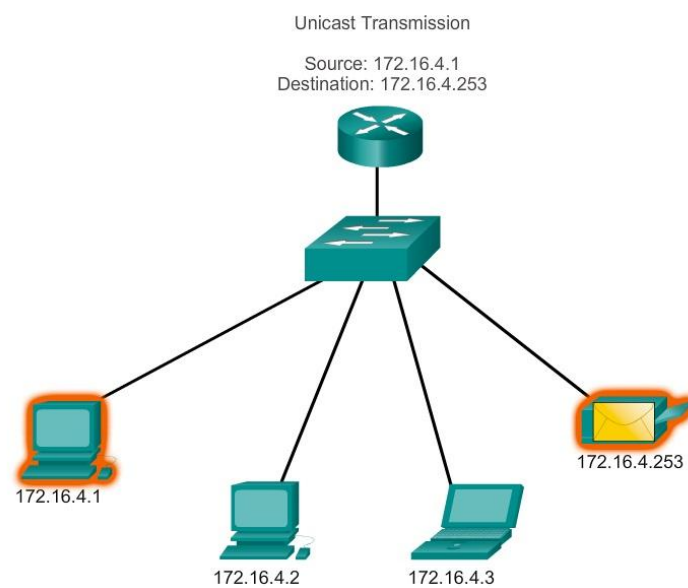
Types of IP address

IP communication commonly uses Unicast, Multicast, Broadcast, and Anycast delivery concepts.

IPv4 supports unicast, multicast, and broadcast communication.

IPv6 supports unicast, multicast, and Anycast communication, but it does not use broadcast addresses.

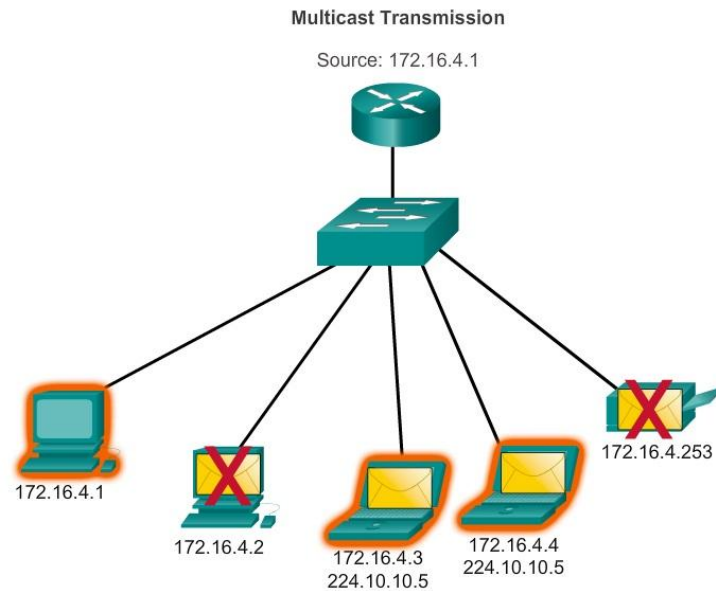
Unicast Address: Identifies uniquely a device on a network. It is used for communication between two devices. A data traffic is directed to a specific network device.



This is a Unicast traffic from PC .4.1 to Printer .4.253

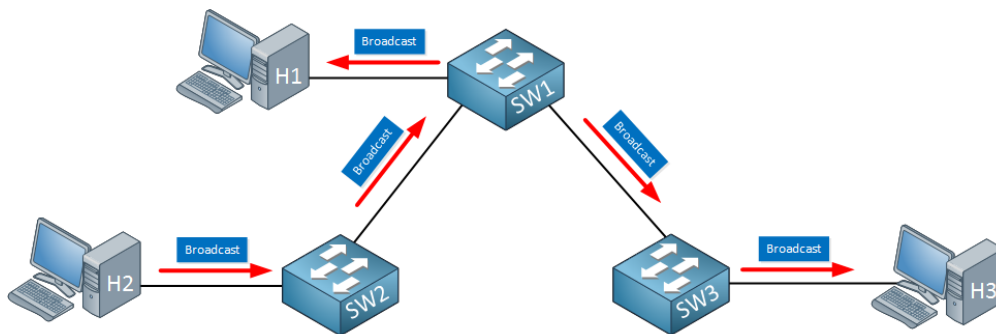
Multicast Address: Is an address that identifies a group of devices. A data traffic is directed to multiple network devices but not to all devices in the network.

Multicast IP Address Range: 224.0.0.0 – 239.255.255.255



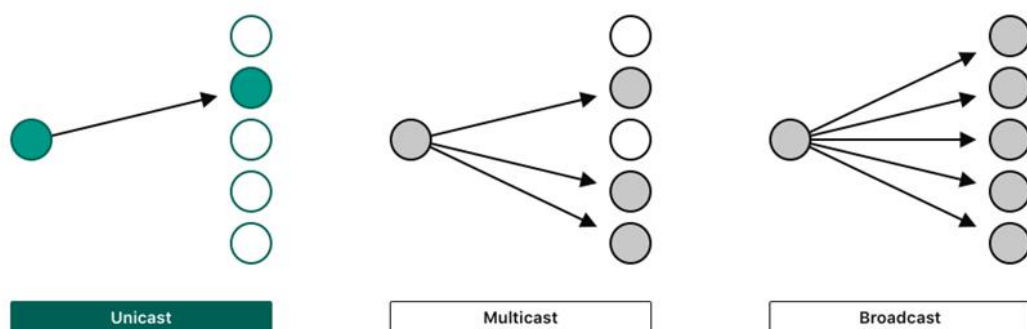
A Multicast traffic from PC .4.1 to PC .4.3 & PC .4.4

Broadcast Address: Is an address that identifies all the devices on a network. A data traffic that is directed to every device on a network.



A broadcast traffic from PC H2 to Broadcast Domain

Unicast	One-to-One Communication
Multicast	One-to-Many Communication
Broadcast	One-to-All Communication



Unicast vs Multicast vs Broadcast

Routing Protocols

Routing protocols are important for defining the best route for data packets to travel across a network. They allow routers to communicate with each other, share information about changes in network topology, and make decisions on the most efficient routes for data transmission.

Interior Gateway Protocols (IGP)

Used in a single Autonomous System (AS), such as RIP, RIPv2, EIGRP, OSPF.

Exterior Gateway Protocols (EGP)

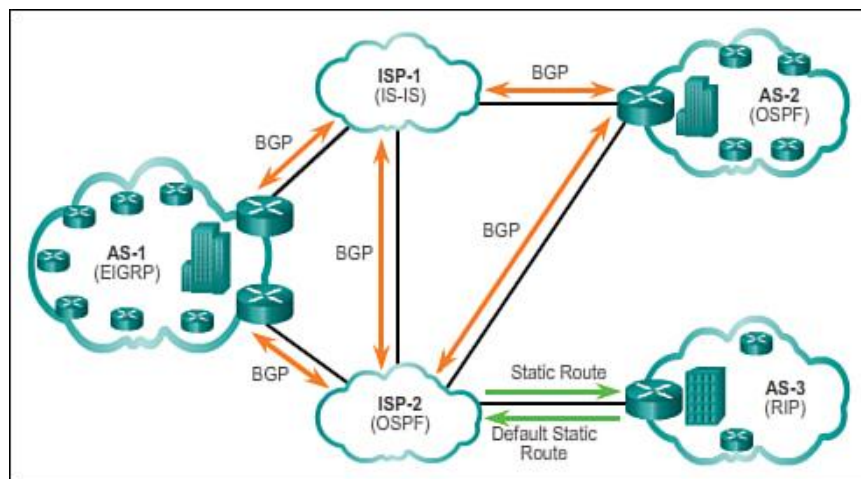
Used to route traffic between different Autonomous Systems, such as: BGP (Boarder Gateway Protocol).

Autonomous System (AS)

An autonomous system is the collection of connected networks that are managed by a single organization. Each AS has a unique number called Autonomous System Number (ASN), which is used by BGP in order to exchange routing information between different ASes.

There are three types of Autonomous Systems:

1. **Stub AS**: Connects to only one AS.
2. **Multi-Homed AS**: Connects to multiple ASes without transit the information.
3. **Transit AS**: Connects Multiple ASes but also allows data to pass through.



Autonomous System

The protocols are divided into *Distance Vector Protocols*, *Link State Protocols*, and *Path Vector Protocols*.

Distance Vector Routing Protocols

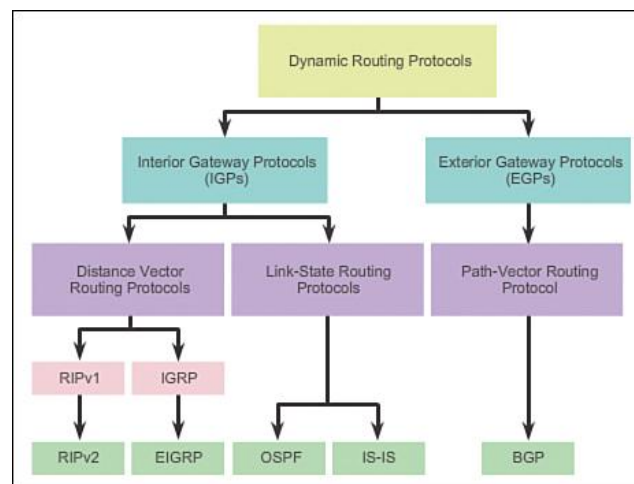
They determine the best path based on hops, i.e. in how many hops will the information have to take to reach the destination. (RIP, IGRP cisco proprietary protocol)

Link State Routing Protocols

Have a complete view of the network topology and use algorithms to calculate the shortest (fastest) path (OSPF, IS-IS).

Path Vector Routing Protocols

Maintain the path information that gets updated dynamically as the network topology changes (BGP).



Dynamic Routing Protocols

Boarder Gateway Protocol (BGP)

The BGP is an exterior gateway protocol which is used to exchange routing information between different Autonomous Systems on the Internet.

The way that the BGP works is as follows:

- Peering: The BGP Routers (peers) establish a connection in order to exchange routing information.
- Route Advertisement: The BGP Routers advertise the routes they know to the other already connected routers.
- Path Selection: The way that BGP chooses the right path is based on many metrics such as path length and policy rules.
- Route Propagation: As soon as a router decides the best path then it propagates the path to the other peers to have the updated routing information.

Layer 4: Transport Layer

The Transport Layer provides end-to-end communication between applications.

Functions such as reliability, ordering, acknowledgements, retransmission, flow control, and connection management depend on the transport protocol. TCP provides these mechanisms, while UDP does not provide delivery guarantees.

TCP Layer basic features:

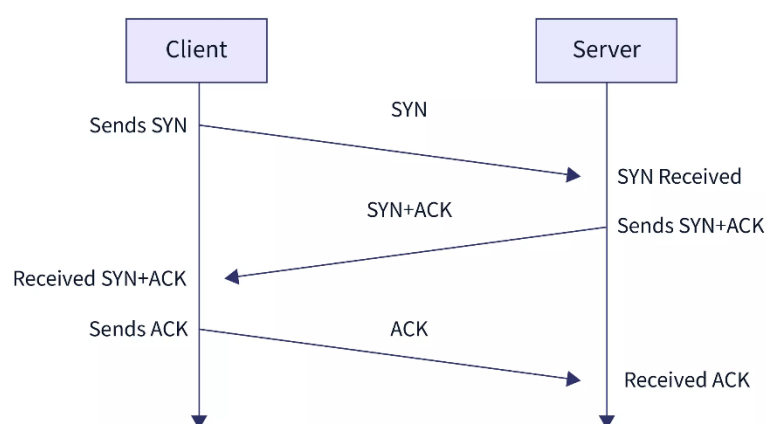
- ◆ End-to-end Communication: Offers connectivity between source and destination device.
- ◆ Segmentation & Reassembly: It breaks big data streams into smaller and easier to manage data segments. At the destination point transport layer is obliged to reassembly the data segments into their original form.
- ◆ Connection Management: It establishes, maintains, and terminates the connection between communicating devices.
- ◆ Flow Control: Controls the data transmission rate to avoid network congestion.
- ◆ Error Detection & Correction: It detects errors and corrupted data must be retransmitted.
- ◆ Multiplexing: Multiple apps can use the network simultaneously.

The TCP Layer has two main transmission protocols:

1. TCP (Transmission Control Protocol)

It checks the status of the connection and also checks whether the data was received correctly by the user. It is Full Duplex, meaning the data exchange takes place at the same time.

Prior to data transmission the TCP must establish a connection between the two ends. This process is called *3-Way Handshake*. It's a fundamental process used in the Transmission Control Protocol (TCP) to establish a reliable connection between a client and a server, in order to exchange data.



3-way Handshake

3-way Handshake Procedure

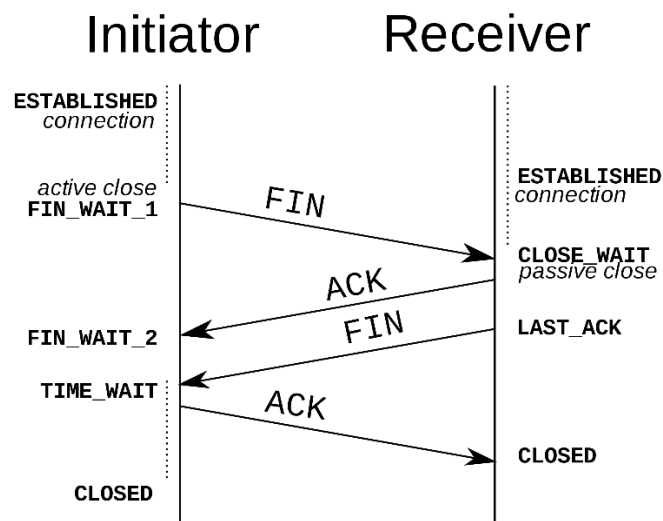
- ◆ SYN packet: The client sends a synchronization packet to the server to initiate a connection.
- ◆ SYN-ACK packet: The server responds to the client with an Acknowledge packet.
- ◆ ACK packet: Finally the client responds to server with an Acknowledge packet and then the connection is established, and the two sides are ready to exchange data.

TCP uses a *sliding-window mechanism* for flow control.

The receiver advertises how much additional data it is prepared to accept. TCP may use cumulative and delayed acknowledgements and does not necessarily send one acknowledgement for every received segment.

Congestion control is a separate mechanism that adjusts transmission behavior according to conditions within the network.

Finally, after the communication is completed and one of the two ends wishes to terminate the connection (client for example), sends a FIN packet, the recipient (server) responds with an ACK packet and at the same time sends another FIN (ACK/FIN) packet and the side that started (the client) it sends a final ACK packet, and the connection is terminated.



TCP Connection Termination Procedure

UDP (User Datagram Protocol)

It is also a protocol of the fourth layer of the OSI Reference model and its purpose is transmission speed and not transmission reliability.

The main features of UDP are:

1. **Connectionless**: Unlike TCP, UDP does not set up communication before sending data. Which makes it fast but not as reliable.
2. **No Guarantees**: It has no mechanism to check that the package was delivered correctly or that it was delivered in the correct order.
3. **Low Overhead**: Because of this it is suitable for applications where speed is essential.
4. **Usage**: It is mainly used in applications that are time sensitive, such as video streaming, VoIP, DNS, Online Gaming.

TCP vs UDP

Key Differences	TCP	UDP
Reliability	Reliable & Ensures Data Integrity	Fast but No Delivery Guarantees.
Connection	Connection Oriented	Connectionless
Overhead	Higher Overhead	Minimal Overhead
Usage	Reliability & Data Integrity	Speed & Low Latency

Layer 5: Session Layer

This layer is responsible for establishing, managing, and terminating sessions between applications.

Key Features

- Session Management: Establishes, maintains, and terminates communication sessions between applications.
- Communication Control: Manages the communication between two devices, ensuring that data is properly synchronized, and the two devices have no problems in communication.
- Data Synchronization: Provides synchronization in data traffic between devices, allowing for recovery and continuation of communication in case of disruptions.

Layer 5 Protocols

The most important protocols in Session Layer are, the NetBIOS and RPC (Remote Procedure Call).

NetBIOS (Network Basic Input Output System): Allows applications on different computers to communicate within a LAN. It permits application to establish sessions, send/receive data and terminate sessions.

It has three basic services:

- ⌘ Name Services: Lists and Resolves NetBIOS names to IP addresses.
- ⌘ Datagram Service: Supports Connectionless Communication.
- ⌘ Session Service: Supports Connection-Oriented Communication.

RPC (Remote Procedure Call): Allows a program to execute a procedure on a remote server as if it were on the same network.

Layer 6: Presentation Layer

This layer is responsible for the translation, encryption, and compression of data. Converts data from the application layer into a format that can be sent over the network. It encrypts data before transmission and decrypts them at the destination, that way data is protected from unauthorized access. In order to improve transmission speed it compress the data in order to reduce the size. A very important part of Layer 6 is to ensure that the data is presented in an understandable way from the receiving application.

Layer 6 Protocols and Standards

TLS (Transport Layer Security): Provides encryption for secure communication.

SSL (Secure Socket Layer): An older protocol for securing data transmission.

MIME (Multipurpose Internet Mail Extensions): Used for formatting multimedia email messages.

Layer 7: Application Layer

The final layer in OSI Reference model stack. Is the layer that is the closest to the final user and ensures that data is presented in a readable format for the user. Moreover, it provides network services directly to the end-user applications.

Application Layer Protocols	
HTTP/HTTPS	For Web Browsing
FTP	For File Transfers
SMTP	For Sending emails
DNS	For Domain name resolution

Application Layer Services	
File Transfer	File Transfer between devices
Email Services	Enables sending/receiving emails
Web Browsing	Allows interaction of users and web apps
Directory Services	Provides access to Directory Databases

Troubleshooting and Monitoring Tools

No matter how well a network is configured, we should always check its correct operation. To achieve this there are many tools either of the operating system we use or of a third party.

In a Windows OS environment we can use the CMD commands in order to identify that our network is working as it should.

CMD Networking Troubleshooting Commands	
ipconfig or ipconfig /all	Checks the TCP network configuration
ipconfig /displaydns	Displays entries in the local Windows DNS resolver cache.
Ipconfig /flushdns	Clears the local DNS resolver cache. It does not delete records from a DNS Server
ping	Sends ICMP Echo requests to test whether replies are received. A failed ping does not always prove that the host is offline
arp -a	Displays ARP entries
tracert	Checks and counts the path to a host
route print	Displays the routing table.
netstat	Displays information about local network connections, listening ports, and protocol statistics, depending on the selected parameters
nslookup	Diagnoses DNS issues
pathping	Combines Tracert & ping

Support Protocols

- ◆ ICMP (Internet Control Message Protocol): ICMP carries control, diagnostic, and error-reporting messages for IP communication. The ping command commonly uses ICMP Echo Request and Echo Reply messages.
- ◆ ARP (Address Resolution Protocol): We know the recipient's IP address and look up its MAC address.
- ◆ RARP (Reverse ARP): *Reverse ARP was an early protocol used by a device to request an IPv4 address based on its hardware address. It has been replaced by more capable protocols such as BOOTP and DHCP.*

Telnet & SSH (Secure Shell): Remote access to a device such as a router, switch, or firewall, via the command line. The main difference is that in the case of SSH the communication is encrypted.